

CLASSIFICATION

COUNTING THE COST – PART II

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COUNTING THE COST

The following concepts will be introduced:

- ✓ **CUMULATIVE GAINS**
- ✓ **LIFT FACTOR**

However, **YOU CAN QUESTION YOUR FRIEND** because in practice, **COSTS ARE RARELY KNOWN WITH ANY DEGREE OF ACCURACY**, and it is known that it is always better to **PONDER VARIOUS DIFFERENT SCENARIOS**.

Your friend remind you that the **PROMOTIONAL ACTION OF THE** Telecom Company **CAN BE OFFERED TO NO MORE THAN 500 CUSTOMERS WHICH MUST BE CAREFULLY SELECTED IN SUCH A WAY THAT MOST OF THEM ARE CHURNERS**.



CUSTOMERS

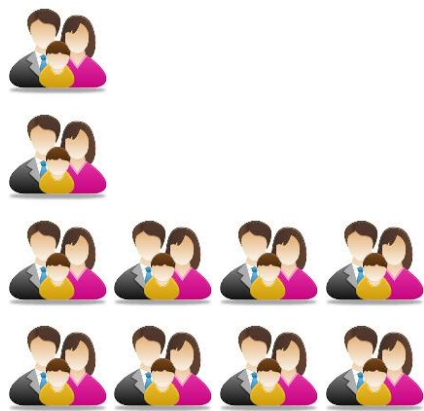
1,334

POTENTIAL CHURNERS

500

However, **YOU CAN QUESTION YOUR FRIEND** because in practice, **COSTS ARE RARELY KNOWN WITH ANY DEGREE OF ACCURACY**, and it is known that it is always better to **PONDER VARIOUS DIFFERENT SCENARIOS**.

Your friend tells you that, based on the experience accumulated through previous promotional actions, **THE CHURNERS ARE 15% OF THE CUSTOMERS (1,334*0.15=200 CHURNERS OUT OF 1,334 CUSTOMERS AND 500*0.15=75 CHURNERS OUT OF 500 POTENTIAL CHURNERS).**



CUSTOMERS
1,334



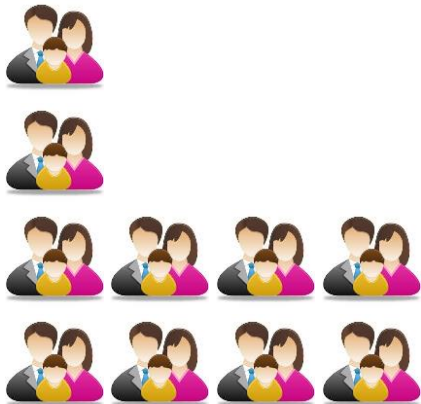
POTENTIAL CHURNERS
500

RANDOM SAMPLING

	-1	+1	
-1	709	425	→ 37.5% 75/(125+75)
+1	125	75	

However, **YOU CAN QUESTION YOUR FRIEND** because in practice, **COSTS ARE RARELY KNOWN WITH ANY DEGREE OF ACCURACY**, and it is known that it is always better to **PONDER VARIOUS DIFFERENT SCENARIOS**.

Your friend also tells you that the **PROPORTION OF CHURNERS** which is known to be **CORRECTLY MAILED USING THE SMP IS 50%** ($200 \times 0.5 = 100$ CHURNERS).



CUSTOMERS
1,334

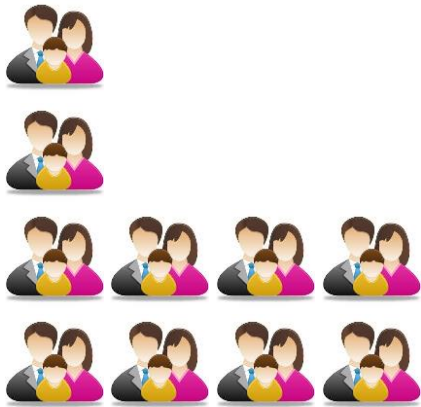


POTENTIAL CHURNERS
500

STANDARD MAILOUT PROCEDURE	-1	+1	→ 50% $100 / (100 + 100)$
	-1 734	400	
	+1 100	100	
RANDOM SAMPLING	-1	+1	→ 37.5% $75 / (125 + 75)$
	-1 709	425	
	+1 125	75	

However, **YOU CAN QUESTION YOUR FRIEND** because in practice, **COSTS ARE RARELY KNOWN WITH ANY DEGREE OF ACCURACY**, and it is known that it is always better to **PONDER VARIOUS DIFFERENT SCENARIOS**.

You developed a **CLASSIFICATION MODEL (M)** that when applied to the set of customers selects a subset of **500 POTENTIAL CHURNERS**, **SUCH THAT 60% OF CHURNERS ARE CORRECTLY IDENTIFIED** ($200 \times 0.6 = 120$ CHURNERS).



CUSTOMERS
1,334

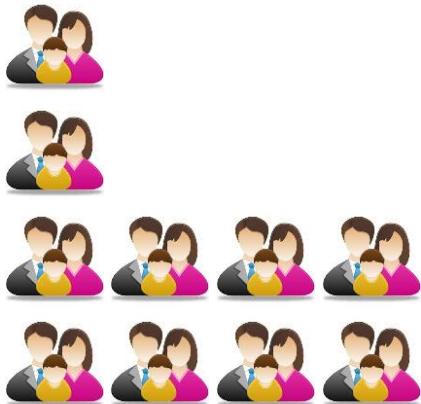


POTENTIAL CHURNERS
500

CLASSIFICATION MODEL M	-1	-1	+1	→ 60% 120 / (80 + 120)
	+1	754 80	380 120	
STANDARD MAILOUT PROCEDURE	-1	-1	+1	→ 50% 100 / (100 + 100)
	+1	734 100	400 100	
RANDOM SAMPLING	-1	-1	+1	→ 37.5% 75 / (125 + 75)
	+1	709 125	425 75	

However, **YOU CAN QUESTION YOUR FRIEND** because in practice, **COSTS ARE RARELY KNOWN WITH ANY DEGREE OF ACCURACY**, and it is known that it is always better to **PONDER VARIOUS DIFFERENT SCENARIOS**.

It may well **PAY OFF TO RESTRICT THE MAILOUT TO THESE 500 CUSTOMERS** – that depends on the **MAILING COST COMPARED TO THE RETURN GAINED FOR EACH RESPONSE TO THE PROMOTION**.



CUSTOMERS
1,334



POTENTIAL CHURNERS
500

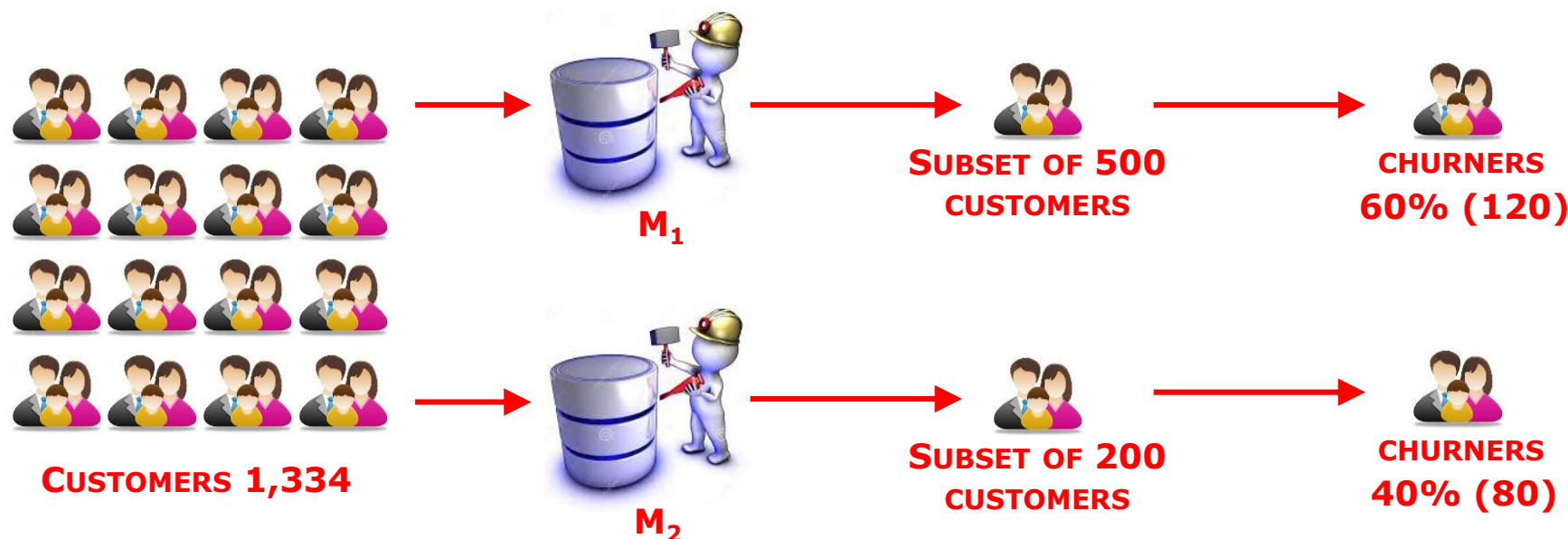
CLASSIFICATION MODEL M	-1	-1	+1	→	60%
	+1	754	380		120/(80+120)
STANDARD MAILOUT PROCEDURE	-1	-1	+1	→	50%
	+1	734	400		100/(100+100)
RANDOM SAMPLING	-1	-1	+1	→	37.5%
	+1	709	425		75/(125+75)



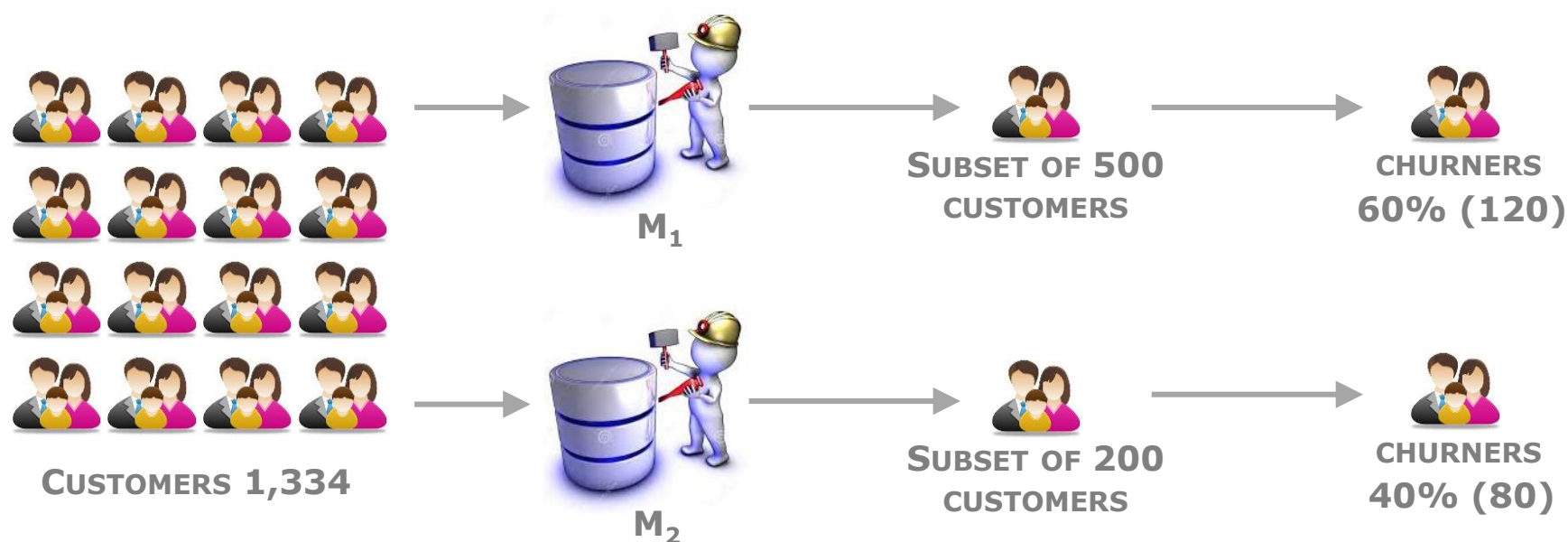
Your friend tells you that in **MARKETING TERMINOLOGY**, the **INCREASE IN RESPONSE RATE**, a factor of 1.6 in this case (37.5% to 60%), is known as the **LIFT FACTOR YIELDED BY THE CLASSIFICATION MODEL (M) WITH RESPECT TO THE RANDOM SAMPLING PROCEDURE**.

Your friend also tells you that **KNOWING THE COST WILL RESULT IN DETERMINING THE PAYOFF IMPLIED BY A PARTICULAR LIFT FACTOR**.

However, **CHANGING THE PARAMETER SETTING OF THE CLASSIFICATION MODEL (M)** can result in **IDENTIFYING DIFFERENT CUSTOMERS**.



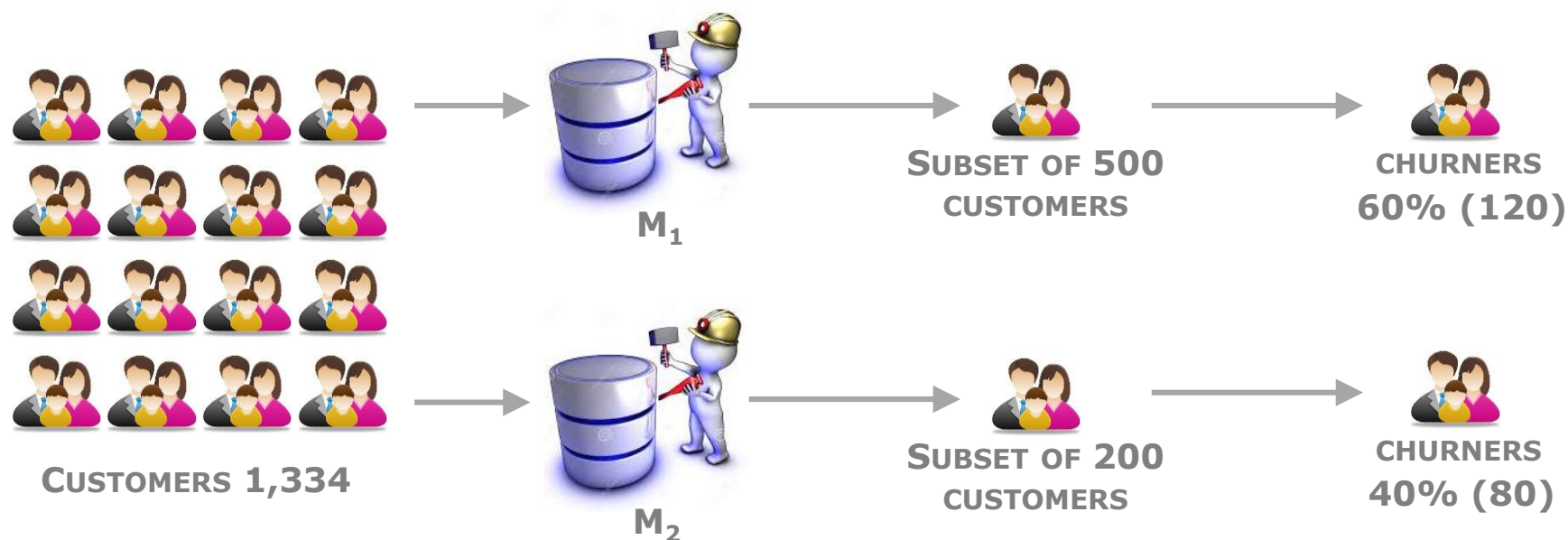
WHICH IS THE MOST PROFITABLE TARGET FOR THE MAILOUT?



WHICH IS THE MOST PROFITABLE TARGET FOR THE MAILOUT?



PROFITABILITY CAN BE COMPUTED ONCE YOU KNOW THE COSTS INVOLVED



Given a **CLASSIFICATION MODEL (M)** which **OUTPUTS THE PREDICTED PROBABILITIES FOR THE POSITIVE CLASS**. You must **FIND SUBSETS OF THE CUSTOMERS SET THAT HAVE:**

- ✓ High proportion of positive records
- ✓ Higher than in the customers set

Assume the set of **CUSTOMERS CONSISTS OF JUST 10 INDIVIDUALS**.

You apply the classification model **M** to this 10 records and get the probability value that each of them is a churner (**Prob(y)**).

RowID	Prob(y)
67	0.47
2,553	0.15
289	0.10
134	0.95
90,034	0.73
874,823	0.64
2,349	0.03
98,324	0.32
18,923	0.86
221	0.88

Given a **CLASSIFICATION MODEL (M)** which **OUTPUTS THE PREDICTED PROBABILITIES FOR THE POSITIVE CLASS**. You must **FIND SUBSETS OF THE CUSTOMERS SET THAT HAVE:**

- ✓ High proportion of positive records
- ✓ Higher than in the customers set

You **SORT** the **customer records of the customers set** in **descending order of the probability** (Prob(y)).

Rank	RowID	Prob(y)
1	134	0.95
2	221	0.88
3	18,923	0.86
4	90,034	0.73
5	874,823	0.64
6	67	0.47
7	98,324	0.32
8	2,553	0.15
9	289	0.10
10	2,349	0.03

Given a **CLASSIFICATION MODEL (M)** which **OUTPUTS THE PREDICTED PROBABILITIES FOR THE POSITIVE CLASS**. You must **FIND SUBSETS OF THE CUSTOMERS SET THAT HAVE:**

- ✓ High proportion of positive records
- ✓ Higher than in the customers set

You **FIND A SUBSET OF A GIVEN SIZE (3) WITH THE GREATEST POSSIBLE PROPORTION OF POSITIVE RECORDS**, you just read the requisite number of records off the list, starting at the top.

Rank	RowID	Prob(y)
1	134	0.95
2	221	0.88
3	18,923	0.86
4	90,034	0.73
5	874,823	0.64
6	67	0.47
7	98,324	0.32
8	2,553	0.15
9	289	0.10
10	2,349	0.03

Compute the **LIFT FACTOR**.

Given a **CLASSIFICATION MODEL (M)** which **OUTPUTS THE PREDICTED PROBABILITIES FOR THE POSITIVE CLASS**. You must **FIND SUBSETS OF THE CUSTOMERS SET THAT HAVE:**

- ✓ High proportion of positive records
- ✓ Higher than in the customers set

You **FIND A SUBSET OF A GIVEN SIZE (3) WITH THE GREATEST POSSIBLE PROPORTION OF POSITIVE RECORDS**, you just read the requisite number of records off the list, starting at the top.

Rank	RowID	Prob(y)	Churn
1	134	0.95	y
2	221	0.88	y
3	18,923	0.86	n
4	90,034	0.73	n
5	874,823	0.64	n
6	67	0.47	n
7	98,324	0.32	n
8	2,553	0.15	y
9	289	0.10	n
10	2,349	0.03	n

Compute the **LIFT FACTOR**.

To compute the **LIFT FACTOR** you need to compute how many churners the classification model M will identify when used to select 3 customers.

Given a **CLASSIFICATION MODEL (M)** which **OUTPUTS THE PREDICTED PROBABILITIES FOR THE POSITIVE CLASS**. You must **FIND SUBSETS OF THE CUSTOMERS SET THAT HAVE:**

- ✓ High proportion of positive records
- ✓ Higher than in the customers set

You **FIND A SUBSET OF A GIVEN SIZE (3) WITH THE GREATEST POSSIBLE PROPORTION OF POSITIVE RECORDS**, you just read the requisite number of records off the list, starting at the top.

Rank	RowID	Prob(y)	Churn
1	134	0.95	y
2	221	0.88	y
3	18,923	0.86	n
4	90,034	0.73	n
5	874,823	0.64	n
6	67	0.47	n
7	98,324	0.32	n
8	2,553	0.15	y
9	289	0.10	n
10	2,349	0.03	n

subset consisting of 3 records

2 records out of the 3 positive ones
are correctly identified

$2/3 = 0.66$ of positive records

Lift = $0.66/0.3 = 2.22$

Given a **CLASSIFICATION MODEL (M)** which **OUTPUTS THE PREDICTED PROBABILITIES FOR THE POSITIVE CLASS**. You must **FIND SUBSETS OF THE CUSTOMERS SET THAT HAVE:**

- ✓ High proportion of positive records
- ✓ Higher than in the customers set

You **FIND A SUBSET OF A GIVEN SIZE (3) WITH THE GREATEST POSSIBLE PROPORTION OF POSITIVE RECORDS**, you just read the requisite number of records off the list, starting at the top.

Rank	RowID	Prob(y)	Churn
1	134	0.95	y
2	221	0.88	y
3	18,923	0.86	n
4	90,034	0.73	n
5	874,823	0.64	n
6	67	0.47	n
7	98,324	0.32	n
8	2,553	0.15	y
9	289	0.10	n
10	2,349	0.03	n

subset consisting of 3 records

2 records out of the 3 positive ones
are correctly identified

$2/3 = 0.66$ of positive records

Lift = $0.66 / 0.3 = 2.22$

0.3 (3/10) is the fraction of churners identified when the random sampling procedure is applied.



Given a **CLASSIFICATION MODEL (M)** which **OUTPUTS THE PREDICTED PROBABILITIES FOR THE POSITIVE CLASS**. You must **FIND SUBSETS OF THE CUSTOMERS SET THAT HAVE:**

- ✓ High proportion of positive records
- ✓ Higher than in the customers set

You **FIND A SUBSET OF A GIVEN SIZE (3)** WITH THE **GREATEST POSSIBLE PROPORTION OF POSITIVE RECORDS**, you just read the requisite number of records off the list, starting at the top.

Rank	RowID	Prob(y)	Churn
1	134	0.95	y
2	221	0.88	y
3	18,923	0.86	n
4	90,034	0.73	n
5	874,823	0.64	n
6	67	0.47	n
7	98,324	0.32	n
8	2,553	0.15	y
9	289	0.10	n
10	2,349	0.03	n

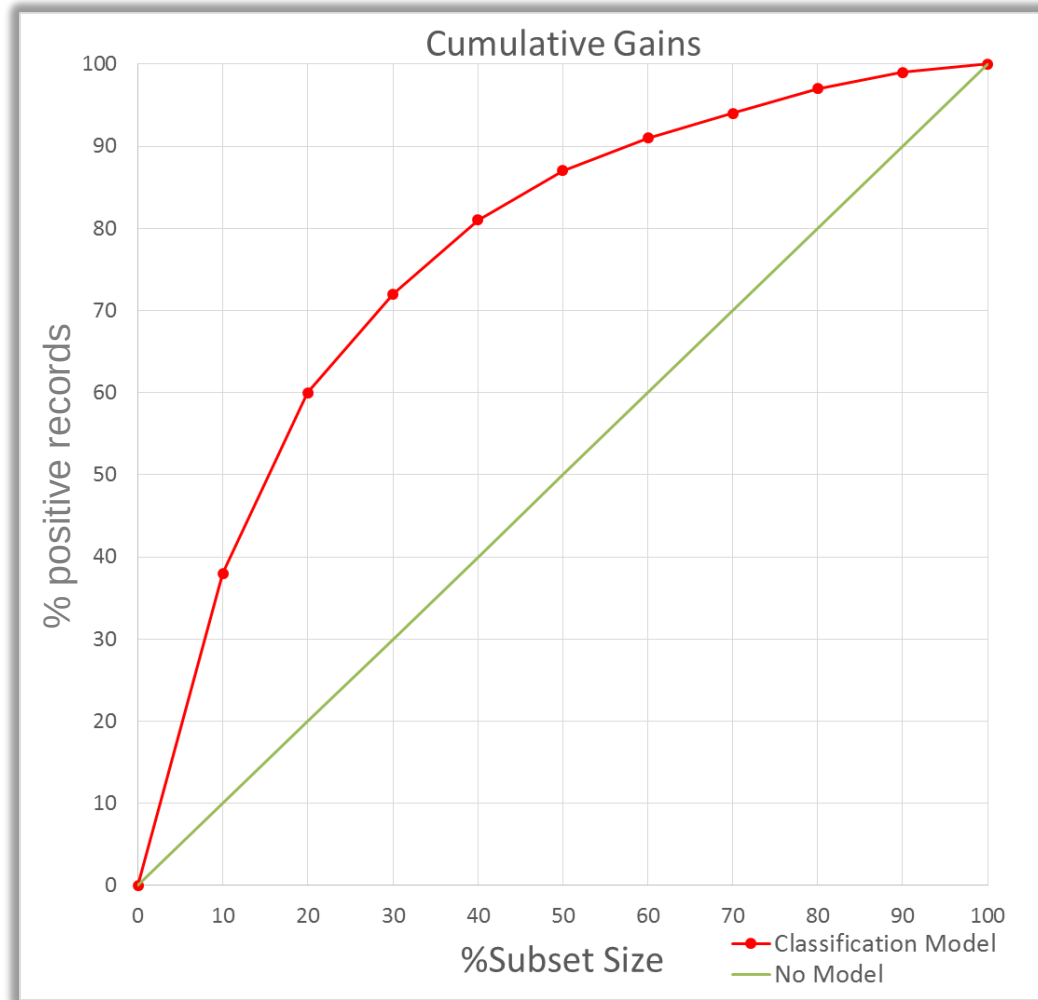
subset consisting of 5 records

2 records out of the 3 positive ones
are correctly identified

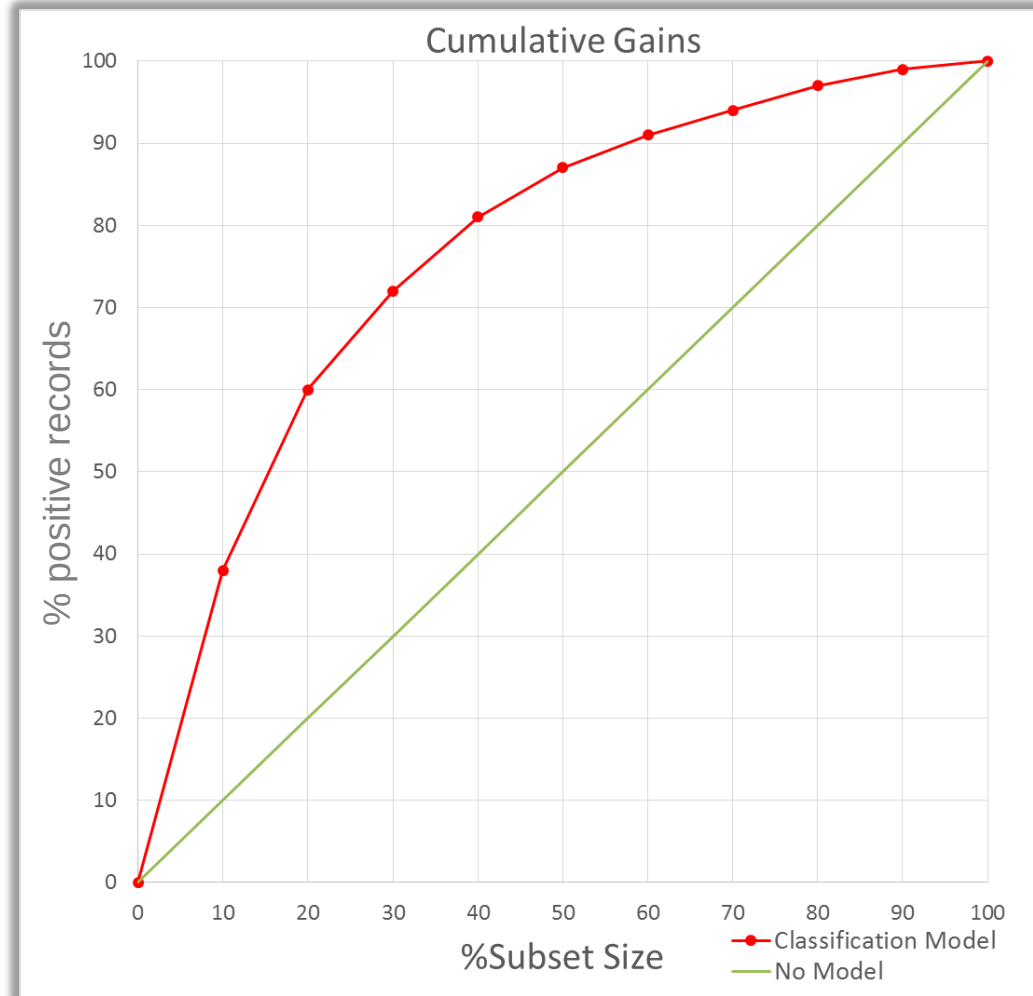
$2/3 = 0.66$ of positive records

Lift = $0.66/0.5 = 1.33$

A graphical representation of the empirical procedure just shown, which allows to have a depiction of the various possibilities, will be far more revealing than presenting a single optimal decision.



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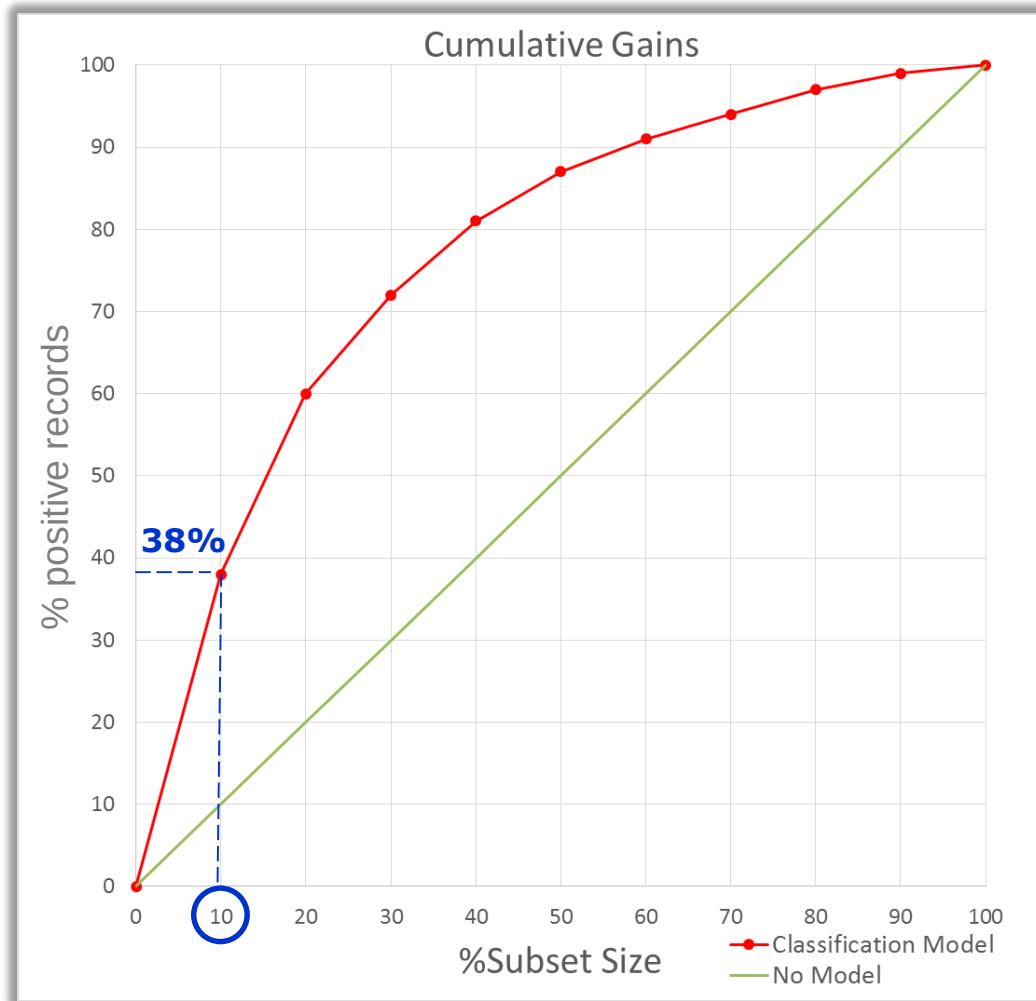


YOU SELECT a given number of customers to mailout, i.e. **1,000 CUSTOMERS**.

COUNTING THE COST: CUMULATIVE GAINS

4

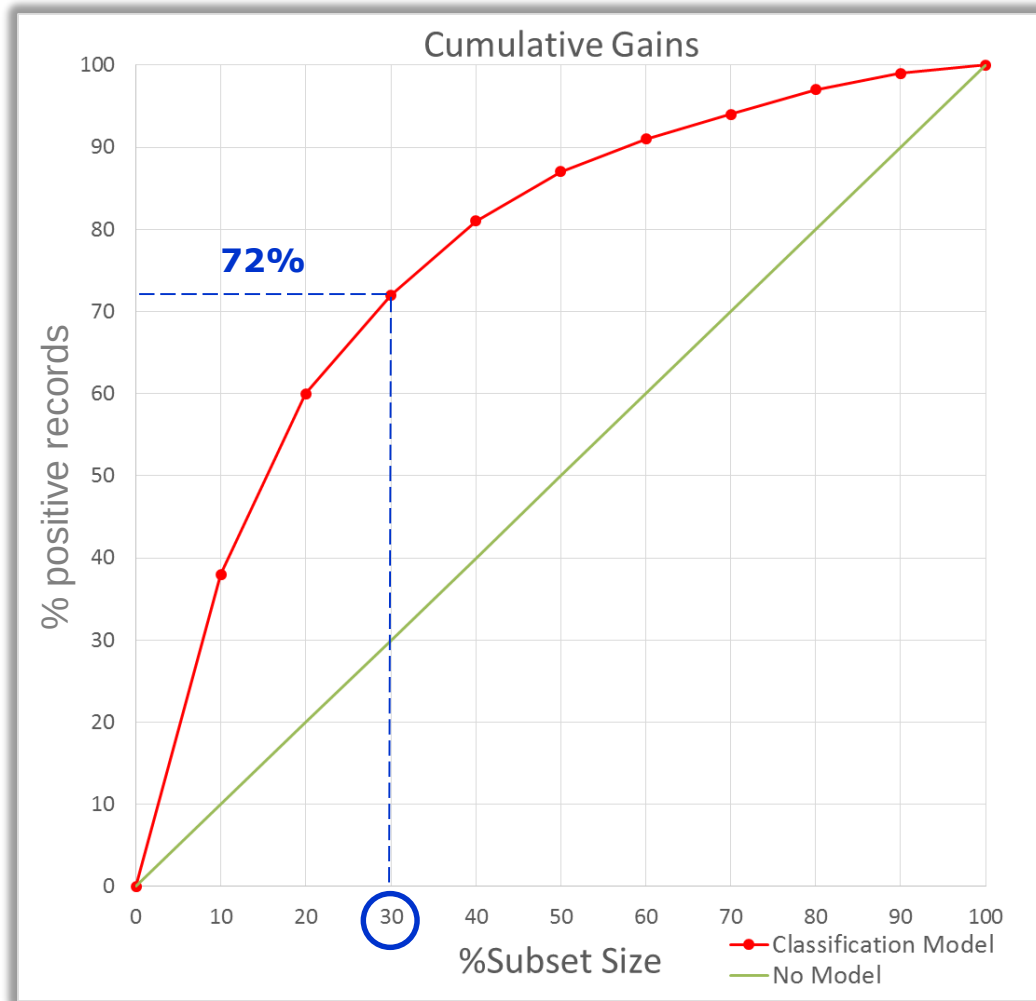
A graphical representation of the empirical procedure just shown, which allows to have a depiction of the various possibilities, will be far more revealing than presenting a single optimal decision.



YOU SELECT a given number of customers to mailout, i.e. **1,000 CUSTOMERS**.

Assume it corresponds to the **10% OF THE CUSTOMERS SET**.

A graphical representation of the empirical procedure just shown, which allows to have a depiction of the various possibilities, will be far more revealing than presenting a single optimal decision.



YOU SELECT a given number of customers to mailout, i.e. **1,000 CUSTOMERS**.

Assume it corresponds to the **10% OF THE CUSTOMERS SET**.

IF YOU SELECT 3,000 CUSTOMERS then it corresponds to the **30% OF THE CUSTOMERS SET**.